

Factor Analysis Methodology

1. Data Preparation

The dataset used in this analysis consists of personality survey responses (Big Five traits). Only relevant questionnaire items were selected, including variables with prefixes: EXT (Extraversion), AGR (Agreeableness), CSN (Conscientiousness), EST (Emotional Stability), and OPN (Openness).

Missing values were removed, and a random sample of 2000 observations was used to ensure computational efficiency while maintaining sufficient statistical power.

```
import pandas as pd
```

```
# Load dataset
```

```
df = pd.read_csv("data-final.csv", sep='\t')
```

```
# Select relevant variables
```

```
df_questions = df[[col for col in df.columns if col[:3] in ["EXT", "EST", "AGR", "CSN", "OPN"]]]
```

```
# Drop missing values and sample
```

```
df_sample = df_questions.dropna().sample(2000, random_state=42)
```

2. Extraction Method

Exploratory Factor Analysis (EFA) was conducted using the **FactorAnalyzer** package in Python.

The number of factors was determined using **eigenvalues greater than 1 (Kaiser criterion)**.

```
from factor_analyzer import FactorAnalyzer
```

```
fa = FactorAnalyzer(rotation=None)
```

```
fa.fit(df_sample)
```

```
# Eigenvalues
```

```
ev, v = fa.get_eigenvalues()
```

3. Rotation

To improve interpretability, **Varimax rotation** was applied. This orthogonal rotation method maximizes the variance of loadings, making factor interpretation clearer.

```
fa = FactorAnalyzer(n_factors=5, rotation='varimax')
```

```
fa.fit(df_sample)

# Loadings
loadings = pd.DataFrame(
    fa.loadings_,
    index=df_sample.columns,
    columns=[f'Factor{i+1}' for i in range(5)]
)
```

4. Interpretation of Factors

Factor loadings were examined, and variables with loadings greater than |0.4| were considered significant contributors.

- **Factor 1** showed strong loadings for EXT variables → *Extraversion*
 - **Factor 2** showed strong loadings for AGR variables → *Agreeableness*
 - **Factor 3** showed strong loadings for OPN variables → *Openness*
 - **Factor 4** showed strong loadings for EST variables → *Emotional Stability*
 - **Factor 5** showed strong loadings for CSN variables → *Conscientiousness*
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5. Conclusion

The factor analysis revealed five distinct latent factors corresponding to the **Big Five personality traits**.

The results confirm that the survey data aligns well with established psychological theory, demonstrating clear clustering of variables into meaningful constructs.
